DIVYANSHU GOYAL

Multimodal Model Training | RL Fine-Tuning | Evaluation
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SUMMARY

Applied scientist at Adobe, passionate about multimodal large language models. Over the past 5 years I have trained, aligned, and evaluated these models in highly subjective domains such as brand marketing and photo editing/retouching. I have contributed my work at CVPR, ICML, and AAAI workshops, hold patents in generative AI space, and write about AI through blogs. I am strong believer in first principle thinking and try to bring that to everything I do.

EXPERIENCE

Applied Scientist – Adobe Inc., San Jose, CA June 2021 – Present

VLM Fine-Tuning with SFT and Reinforcement Learning (Gemma-3-12B)

- Trained Gemma-3-12B (LoRA rank 256, 1.19B trainable parameters) to assess marketing-image quality with distributed **SFT**, then multi-round **GRPO**, on 8× H100. Each round samples a group of completions per prompt, scores them with programmatic rewards against ground truth, and updates the policy with a clipped policy-gradient loss. Diagnosed **reward hacking** (collapse onto hedged, low-information answers) and redesigned the reward scorers.

Multimodal Judge Models

- Built a 1.1M-row training set from 14 human-rated sources with rater disagreement as soft labels, and trained **Qwen3.5-4B** LoRA judges on mixes sampled from it. After calibrating its score output on crowd ratings, the served judge raised validation SRCC on KonIQ from 0.421 (reference model) to **0.663** (paired 95% CI above 0), measured on a leak-checked 67K-item held-out **benchmark suite** I built.

Unsupervised Photo Retouching (ATune)

- Led **ATune** (co-first author, under review): a 54M-parameter model that learns photo retouching from 10M unedited photographs with no expert edits and operates off-the-shelf editors by setting their sliders. With 157 raters, its edits beat GPT-6 Astra using the same editor (**54.3%** of decisive votes, 95% CI 50.3–58.3; 3,524 answers) and tied with Claude Opus 5 (50.5%, CI 46.8–54.2). On FiveK, PPR10K, and SPAQ, at least **8 of 9** published quality judges prefer it to GPT-6 Astra.

Evaluation Research

- Created **DistortBench** (first author; CVPR 2026 and ICML 2026 Workshops): 13.5K questions over 27 distortion types and 5 severity levels. Of 18 VLMs, the best (Qwen3.5-27B) scores **61.9%** (chance 25%), above the average annotator (60.2%) and below a 3-annotator majority vote (65.7%).
- Co-authored **JudgeBreak** (ICML 2026 Workshop): linear probes on Qwen3-32B hidden states reach **78.1%** pairwise accuracy across 95 subjective settings, vs. 70.1% for the best reward model or LLM judge.

Product

- Prototyped **Adobe GenStudio for Performance Marketing** (launched October 2024, serving Fortune 500 clients) and built a multimodal **Photoshop agent** that edits images with vision grounding.

Machine Learning Engineer – Adobe Inc., Bangalore, India July 2016 – August 2019

- Co-developed Adobe's production **ML prediction service** (3,500 QPS/node at 0.3–0.6 ms p99; Adobe Tech Blog) and its model lifecycle platform.

PUBLICATIONS

DistortBench: Benchmarking Vision Language Models on Image Distortion Identification
D. Goyal, A. Eppa, V. Bannihatti Kumar | CVPR 2026 and ICML 2026 Workshops | arXiv:2604.19966

JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** | ICML 2026 Workshop | OpenReview

Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari | AAAI 2026 Workshop, Oral | arXiv:2510.05135

PROJECTS

VibeNanoChat: GPT-2 Scale LLM Pretraining from Scratch 2026 | code
Decoder-only transformer (RoPE, RMSNorm, sliding-window attention) with one-knob depth scaling and a custom distributed **Muon** optimizer (ZeRO-2 state sharding), trained in BF16 with FlashAttention-3 at over 1M tokens/sec on 8× H100. Fit isoFLOP curves across 40+ runs (44M to 522M parameters) to find compute-optimal model sizes. Write-ups: Going Fast and A Lazy Engineer's Guide to Scaling Transformers.

PATENTS

Guiding Language Translation with Translation Documents Using ML (granted) 2026 | US20260050753

Brand-Aligned Marketing Content Generation 2025 | US20250117829

Cultural Adaptation of Images Using Generative AI 2026 | US20260051084

Parameter-Efficient Fine-Tuned Models 2025 | US20250217193

TECHNICAL SKILLS

Research: VLM Training & Evaluation, LLM Pretraining & Scaling Laws, Benchmark Design, Human Preference Studies, LLM-as-a-Judge & Reward Models, Reinforcement Learning (GRPO), Reward Design, Linear Probing, Bootstrap Statistics

Tools: Python, PyTorch, Distributed Training (DDP, ZeRO-2), torch.compile, FlashAttention-3, Liger Kernels, Muon, Mixed Precision (BF16), LoRA, Hugging Face Transformers/PEFT, Weights & Biases

EDUCATION

Georgia Institute of Technology – M.S. Computer Science (Machine Learning), GPA 4.0 (2019 – 2021)

BITS Pilani – B.E. Computer Science, M.Sc. Mathematics, GPA 8.75/10 (2011 – 2016)